

15. A 1500 kW, 550-V, 16 pole generator runs at 150 rev. per min. What must be the useful flux if there are 2500 conductors lap-connected and the full-load copper losses are 25 kW? Calculate the area of the pole shoe if the gap density has a uniform value of 0.9 wb/m^2 and find the no-load terminal voltage, neglecting armature reaction and change in speed.

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OBJECTIVE TESTS - 26

- The basic requirement of a d.c. armature winding is that it must be
 - a closed one
 - a lap winding
 - a wave winding
 - either (b) or (c)
- A wave winding must go at least around the armature before it closes back where it started.
 - once
 - twice
 - thrice
 - four times
- The d.c. armature winding in which coil sides are a pole pitch apart is called winding.
 - multiplex
 - fractional-pitch
 - full-pitch
 - pole-pitch
- For making coil span equal to a pole pitch in the armature winding of a d.c. generator, the back pitch of the winding must equal the number of
 - commutator bars per pole
 - winding elements
 - armature conductors per path
 - armature parallel paths.
- The primary reason for making the coil span of a d.c. armature winding equal to a pole pitch is to
 - obtain a coil span of 180° (electrical)
 - ensure the addition of e.m.fs. of consecutive turns
 - distribute the winding uniformly under different poles
 - obtain a full-pitch winding.
- In a 4-pole, 35 slot d.c. armature, 180 electrical-degree coil span will be obtained when coils occupy slots.
 - 1 and 10
 - 1 and 9
 - 2 and 11
 - 3 and 12
- The armature of a d.c. generator has a 2-layer lap-winding housed in 72 slots with six conductors/slot. What is the minimum number of commutator bars required for the armature?
 - 72
 - 432
 - 216
 - 36
- The sole purpose of a commutator in a d.c. Generator is to
 - increase output voltage
 - reduce sparking at brushes
 - provide smoother output
 - convert the induced a.c. into d.c.
- For a 4-pole, 2-layer, d.c., lap-winding with 20 slots and one conductor per layer, the number of commutator bars is
 - 80
 - 20
 - 40
 - 160
- A 4-pole, 12-slot lap-wound d.c. armature has two coil-sides/slot. Assuming single turn coils and progressive winding, the back pitch would be
 - 5
 - 7
 - 3
 - 6
- If in the case of a certain d.c. armature, the number of commutator segments is found either one less or more than the number of slots, the armature must be having a simplex winding.
 - wave
 - lap
 - frog leg
 - multielement
- Lap winding is suitable for current, voltage d.c. generators.
 - high, low
 - low, high
 - low, low
 - high, high

13. The series field of a short-shunt d.c. generator is excited by currents.
 (a) shunt
 (b) armature
 (c) load
 (d) external
14. In a d.c. generator, the generated e.m.f. is directly proportional to the
 (a) field current
 (b) pole flux
 (c) number of armature parallel paths
 (d) number of dummy coils
15. In a 12-pole triplex lap-wound d.c. armature, each conductor can carry a current of 100 A. The rated current of this armature is ampere.
 (a) 600
 (b) 1200
 (c) 2400
 (d) 3600
16. The commercial efficiency of a shunt generator is maximum when its variable loss equals loss.
 (a) constant
 (b) stray
 (c) iron
 (d) friction and windage
17. In small d.c. machines, armature slots are sometimes not made axial but are skewed. Though skewing makes winding a little more difficult, yet it results in
 (a) quieter operation
 (b) slight decrease in losses
 (c) saving of copper
 (d) both (a) and (b)
18. The critical resistance of the d.c. generator is the resistance of
 (a) armature
 (b) field
 (c) load
 (d) brushes (Grad. I.E.T.E Dec. 1985)

ANSWERS

1. (a) 2. (b) 3. (c) 4. (a) 5. (b) 6. (b) 7. (c) 8. (d) 9. (b) 10. (b)
 11. (a) 12. (a) 13. (c) 14. (b) 15. (d) 16. (a) 17. (d) 18. (b)